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Original Research Article

Capsular typing of *Streptococcus pneumoniae* isolated from clinical specimens in Gauhati Medical College and hospital, Assam, India

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ABSTRACT

Purpose: *Streptococcus pneumoniae* is an important human respiratory tract pathogen causing pneumococcal diseases in majority of children and adults. The capsule is a significant virulence factor of *Pneumococci* which determines the bacterial serotype and is the component used for synthesis of pneumococcal vaccines. This cross-sectional study aimed to isolate *Streptococcus pneumoniae* from clinical samples and determine the occurrence of its circulating serotypes in Assam, North East India.

Materials and methods: A total of 80 clinical samples were collected from June 2019 to May 2020 from patients clinically suspected from pneumococcal infection and also included samples routinely sent to bacteriology laboratory. Isolation and identification of *S. pneumoniae* was performed using conventional culture and molecular methods. Antibiotic susceptibility patterns were monitored. Capsular serotyping was performed using PCR of *cpsA* gene followed by DNA sequencing.

Results: Majority of the cases suspected of pneumococcal infection belong to the paediatric group aged less than 5 years. Out of 80 samples, 10 (12.50%) were found to be positive by PCR of *recP* gene. Culture was positive in 80% (8/10) of the total positives. Co-trimoxazole resistance was seen in 33.33% of the isolate from sputum. Serotypes 6A, 6B, 6C and 19F were detected in our region, out of which 6C is a non-vaccine serotype.

Conclusion: Continued surveillance is needed to monitor trends in non-vaccine serotypes that may emerge as highly associated with antibiotic resistance. Also, the need to continuous monitoring of the antibiotic susceptibility of *S. pneumoniae* in North eastern parts of India is of utmost importance.

1. Introduction

Streptococcus pneumoniae is a Gram positive, alpha haemolytic and capsulated bacterium that colonizes the upper respiratory tract of healthy individuals and is one of the most important etiological agents causing life-threatening infections which include Meningitis, Otitis media, Community-acquired pneumonia, Septicaemia etc. In industrialized as well as in developing countries. According to the World Health Organization, nearly 1 million children die of pneumococcal disease annually, with most fatalities occurring in children and the elderly, the colonization being mostly asymptomatic [1,2].

The polysaccharide capsule in *S. pneumoniae* which determines the bacterial serotype is the most important virulence factor of the bacterium. More than 90 serotypes of *Streptococcus pneumoniae* have been

identified till date. But only a fraction of these is responsible for approximately 75% of cases of Invasive Pneumococcal Disease (IPD) worldwide [3]. The worldwide variation in the prevalence of pneumococcal serotypes and drug resistance is related to disparities in the use and misuse of antimicrobial agents and the implementation of pneumococcal vaccination programmes. In India, currently implemented PCV-13 vaccine is expected to provide coverage against 74% of serotypes and also reduce antimicrobial resistance [4,5] However, the prevalence of IPDs and drug resistance by non-vaccine serotypes has increased [6] and increased use of vaccine leads to serotype replacement, rather than reducing its overall burden. Therefore, epidemiological surveillance as well as mapping of regional serotypes is important to understand the population dynamics of *S. pneumoniae* and identify serotype shifts [7]. Several DNA based typing methods targeting the Capsular

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Polysaccharide Synthesis (cps) locus which is the outermost layer of the bacterial cell and the main virulence factor, are being developed [8].

Although several studies have been conducted in the South Indian states, little data is available regarding the occurrence and serotypes of *Streptococcus pneumoniae* in Assam. Therefore, in view of the current scenario, this study was undertaken to determine the occurrence of the *Streptococcus pneumoniae* and its circulating serotypes in Assam, North East India.

2. Materials and methods

2.1. Study design and sample collection

The present study was a cross sectional, hospital based study conducted from June 2019 to May 2020. A total of 80 consecutive clinical samples including Throat swab, Nasopharyngeal swab, Sputum, Pleural fluid, CSF, Aural swab which was routinely sent to the Bacteriology laboratory of our hospital of patients suspected of pneumococcal infection (eg., Pyogenic meningitis, pneumonia, otitis media) was included in the study. The patients were grouped into four age groups 0–20 years, 21–40 years, 41–60 years and >60 years. The samples were cultured on blood agar and chocolate agar medium. The culture plates were incubated at 37 °C with 5% CO₂ by using a candle jar. All pure isolates of *S. pneumoniae* were immediately processed for DNA extraction.

All cases were included after obtaining written consent from them or their guardians (in case of minors). The study was approved by the Institutional Ethics Committee (Human) of our institution prior to initiation of the work and was carried out according to the guidelines of the Committee vide letter No. MC/190/2007/Pt-11/MAR-2019/P.G/126 dated 10-04-2019.

2.2. Culturing conditions

The samples were cultured on Blood agar and Chocolate agar medium and incubated at 37 °C with 5% CO₂ for 18–24 h by using a candle jar. All pure culture isolates of *S. pneumoniae* were stored in Skim-milk-tryptone-glucose-glycerine storage medium at –70 °C. The remaining body fluids (CSF, pleural fluid etc.) were stored at –80 °C for future use. Those plates without any growth were re-incubated for next 24 h. Alpha haemolytic colonies with gram positive lanceolate shaped diplococci were sub-cultured onto a fresh blood agar plate with an optochin disc and were incubated at 37 °C with 5% CO₂ for another 18–24 h. Isolates that exhibit an inhibition zone ≥14 mm around the optochin disc were identified as *S. pneumoniae* and were further confirmed by the bile solubility test.

2.3. Antimicrobial susceptibility testing

Antimicrobial susceptibility testing was performed by the modified Kirby-Bauer's disc diffusion method as per the CLSI 2019 guidelines for the following antibiotics: Erythromycin (15µg), Tetracycline/Doxycycline (30µg), Clindamycin (2µg), Chloramphenicol (30µg), Linezolid (30µg), Vancomycin (30µg), Cotrimoxazole (25 µg). Quality control was performed by using ATCC strain 6303 of *Streptococcus pneumoniae*.

2.4. DNA extraction

Genomic DNA was extracted from all the samples using commercially available QIAamp DNA Mini kit (QIAGEN, Hilden, Germany) as per the manufacturer's instructions and stored at –20 °C.

2.4.1. Detection of *RecP* gene and *CpsA* gene by polymerase chain reaction

All the 80 samples collected were subjected to PCR analysis of transketolase (*RecP*), a housekeeping gene of *S. pneumoniae* for identification of the bacterium and *CpsA* gene (internal control, conserved locus of the bacteria) for serotype identification. PCR analysis of *RecP* and *CpsA* gene was done by conventional PCR amplification method using published

sequences of *S. pneumoniae* specific oligonucleotide primers [9,10] *RecP*-F, (5'-GCCAACTCAGGTATCCAGG-3'), *RecP*-R (5'-TGCAACCGTAGCATT GTAAC-3') and *CpsA*-F (5'-GCAGTACAGCAGTTTGTG GACTGAAC-3'), *CpsA*-R (5'-GAATATTTTCATTATCAGTCCCAGTC-3') to detect a product size of 563 bp for *recP* gene and 160 bp for *CpsA* gene.

Conventional PCR analysis was performed in a final reaction volume of 25 µl containing PCR buffer, 1.5 mM MgCl₂, 200 µM (each) dNTPs, 2U Taq polymerase, 10 µM each of forward and reverse primers and 10–50 ng of genomic DNA as template. Thermal cycling conditions were as follows: initial denaturation at 94 °C for 10 min, followed by 35 cycles of denaturation-30 s at 94 °C, 30 s of annealing at 54 °C (for both *recP* and *cpsA* gene PCR) and 30 s of extension at 72 °C. Final extension was continued for 8 min at 72 °C. In all the PCR assays, genomic DNA from ATCC strain 6303 of *Streptococcus pneumoniae* was used as positive control. The amplified products were visualised by running on 1% agarose gel (w/v).

2.5. DNA sequencing and capsular typing of *CpsA* locus

Out of 10 positive samples, 6 samples (from CSF of patients with Bacterial Meningitis aged <5yrs) were selected for DNA sequencing for identification of serotype by targeting the *cpsA* locus using gene specific primers. DNA sequencing was carried out using Big Dye terminator V 3.1 cycle sequencing ready reaction kit (ABI, Foster City, Ca, USA). Sequences were analyzed in an ABI Prism 3130 Genetic Analyzer (Applied Biosystems, Massachusetts, USA). Raw sequence data were edited using Bioedit version 7.0 (Hall 1999). Further, BLAST analysis was done to find the best possible homolog of our samples.

2.6. Statistical analysis

Data analysis was done using SPSS (version21.0; SPSS Inc, Chicago < IL). The data was denoted as number and percentage was taken as appropriate. For categorical data, comparison was done by using the Chi-square test or fisher's exact test (with Yate's correction wherever required). A p-value <0.05 implies statistical significance of this study.

3. Results

The present study reported a significant number of clinically suspected *S. pneumoniae* cases in the month of November–January in the year 2019–2020, out of which highest percentage of the positive samples was constituted by CSF (80%), followed by sputum (10%) and pleural fluid (10%).

An increased rate of *S. pneumoniae* infection was seen in cases belonging to the male population of our study and in the age group of 0–20 years where 80% positive cases were found in children below 6 yrs of age (8/10). Among adult patients, a single positive infection was seen in each of the age group of 21–40 years and >60 years. However, there was no significant association between the age and sex of patients (p > 0.05) (Table 1).

Rate of culture positivity in our study was found to be 80% (8/10) which was confirmed by optochin sensitivity and bile solubility test. Culture negativity of the remaining two PCR positive samples indicated the molecular method to be more sensitive and accurate than the culture method. Out of 8 culture positives, 6 isolates were from CSF which showed 100% sensitivity to Ceftriaxone, Cefotaxime, Cefepime, Meropenem, Vancomycin, Linezolid and Imipenem. Two isolates from sputum showed 33.33% resistant to co-trimoxazole and 100% sensitivity to Erythromycin, Doxycycline, Clindamycin, Linezolid, Vancomycin and Chloramphenicol (Table 2).

Out of the 80 cases studied, the overall positivity of *S. pneumoniae* by PCR of *recP* gene (Fig. 1a and b) was 12.50% (10/80).

3.1. Serotype identification

Sequence analysis of a total of 6 positive CSF samples revealed serotypes 6A, 6B, 6C and 19F to be most prevalent in our study, out of

Table 1

Age and sex distribution of the cases.

Parameters		No. of Suspected cases (%) (n = 80)	No. of positive cases (%) (n = 10)	Chi sq(df)	p- value
Age group (yrs) (Mean age 17.94 ± 27)	0–20 yrs (< 6years)	53 (66.25%)	8 (80%)	0.9664(1)	0.3559
	21–40 yrs	17 (33.75%)	1 (10%)		
	41–60 yrs	0	0		
	> 60 yrs	10 (12.5%)	1 (10%)		
	Total	80	10		
Sex	Male	49 (61.25%)	7 (70%)	0.368(1)	0.543
	Female	31 (38.75%)	3 (30%)		
	Total	80	10		

Table 2

Antibiogram of the isolates of CSF and sputum.

Sample Type	Antibiotic	Total Culture Positive (CSF: n = 6, Sputum: n = 2)	
		Sensitive (%)	Resistant (%)
CSF	Ceftriaxone	100	0
	Cefotaxime	100	0
	Cefepime	100	0
	Meropenem	100	0
	Vancomycin	100	0
	Linezolid	100	0
Sputum	Imipenem	100	0
	Erythromycin	100	0
	Doxycycline	100	0
	Trimethoprim	66.67	33.33
	-Sulphamethoxazole		
	Clindamycin	100	0
	Linezolid	100	0
	Vancomycin	100	0
	Chloramphenicol	100	0

which serotypes 6A, 6B and 19F were included in PCV13 vaccine whereas serotype 6C was identified to be a non-vaccine serotype (NVT). Upon BLAST analysis, 4 samples showed 97.75% homology with serotype 6A, 6B, 6C whereas the remaining 2 samples showed 100% homology with serotype 6A, 6B, 6C and 19F.

4. Discussion and conclusion

S. pneumoniae is the major causative agent of most of pneumococcal diseases worldwide, where children are most affected and therefore there is an urgent need to gain accurate understanding of the epidemiological aspects of this pathogenic bacterium. Our cross sectional study aimed to demonstrate the occurrence of invasive *S. pneumoniae* infection and its circulating serotypes in Assam, North Eastern part of India which may help in monitoring and identifying new emergent serotypes as well as

implementing the correct treatment regimens. Although the capsule of *S. pneumoniae* is the basis for categorization of serotype, the behaviour and pathogenic potential of a serotype may also be related to the genetic origin of the strain. Therefore molecular typing is of considerable interest [11]. The serotypes 6A, 6B and 6C and 19F were identified in our region in paediatric group less than 5 years, out of which 6C is a non-vaccine serotype which was first described by Park et al., 2007 [12]. These findings are consistent with studies done in other parts of India with some variations [13–15].

The distribution of serotypes varies with geographical location. According to the Pneumococcal global serotype project, serotype 6 and 19 are among the common serotypes present in most Asian countries and are responsible for more than 60% of IPD cases [16]. An accurate determination of *S. pneumoniae* serotypes in a particular region is very crucial and is a key factor, as circulating serotypes gives an idea regarding the vaccination coverage in a particular region and also the pneumococcal diseases caused by a particular serotype. In our study, although serotype determination was done in a limited number of samples, serotype 6C was found to be prevalent in all 6 of the clinical samples sequenced. This non-vaccine serotype was also reported in various other studies [17–19] and may indicate that although a decreasing trend is observed in case of incidence of infections caused by vaccine serotypes, infections involving non-vaccine serotype 6C may be predominantly circulating in Assam, which may favour a high rate of infection risk with this particular variant of *S. pneumoniae*. Therefore surveillance data on distribution of both vaccine and non-vaccine serotypes is of utmost importance in order to facilitate prediction of future trends of serotype distribution for implementation of definite vaccination strategies.

A major limiting factor of our study is the small number of clinical samples analyzed since all the samples were collected from a single hospital. Analysis of more clinical samples would provide better insights on the aetiology of *S. pneumoniae* infection along with possible virulent *S. pneumoniae* strains that might exist in the study population.

In conclusion, pneumococcal infections were more common among children than adults, more common in males than their female counterparts, with seasonal trend observed in most cases diagnosed during the

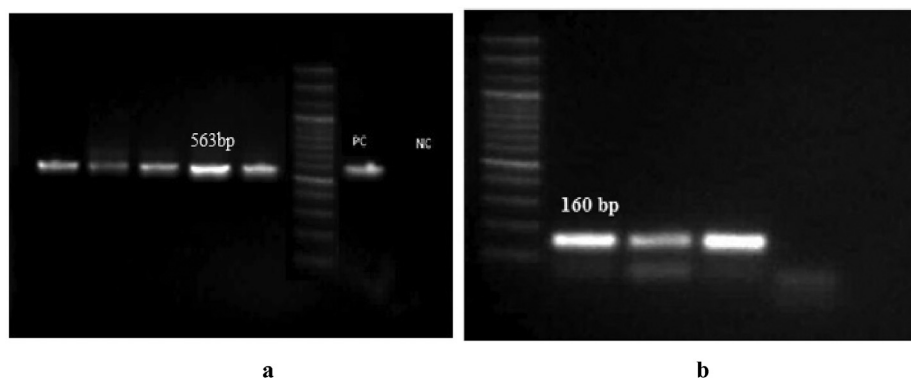


Fig. 1. a: Amplification of *recP* gene (Lane 1–5; 563bp; Lane 6: DNA Ladder 100bp). b: Amplification of *cpsA* gene (Lane 1: DNA Ladder 100bp; Lane 2–4; 160bp) in *S. pneumoniae* positive samples.

winter seasons. Evidences from various studies in India and around the globe suggest that vaccination in children and elderly can reduce the morbidity and mortality due to the diseases caused by *S. pneumoniae*. Continued surveillance is needed to monitor trends in non-vaccine serotypes that may emerge as highly associated with antibiotic non-susceptibility. Also, the need to continuous monitoring of the antibiotic susceptibility of *S. pneumoniae* in North eastern parts of India is of outmost importance.

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Ethical approval

The study was approved by the Institutional Ethics Committee (Human) of our institution vide Letter No. MC/190/2007/Pt-11/MAR-2019/P.G/126 dated 10-04-2019.

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Conflict of interest

None declared.

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